

PROJECT EMBER

Engineering Case Study

NOVEMBER 2026

01 / OVERVIEW

A game and its supporting platform

Original development, ongoing production; selected outcomes demonstrated in engine.



AUBURN STREET / REAL IN-ENGINE STUDY IMAGE

THE GOAL

Auburn Ascendance is an original narrative-driven game. Project Ember is the Unreal Engine C++ project and engineering environment developed to support its gameplay, environments, and iterative production.

MY ROLE

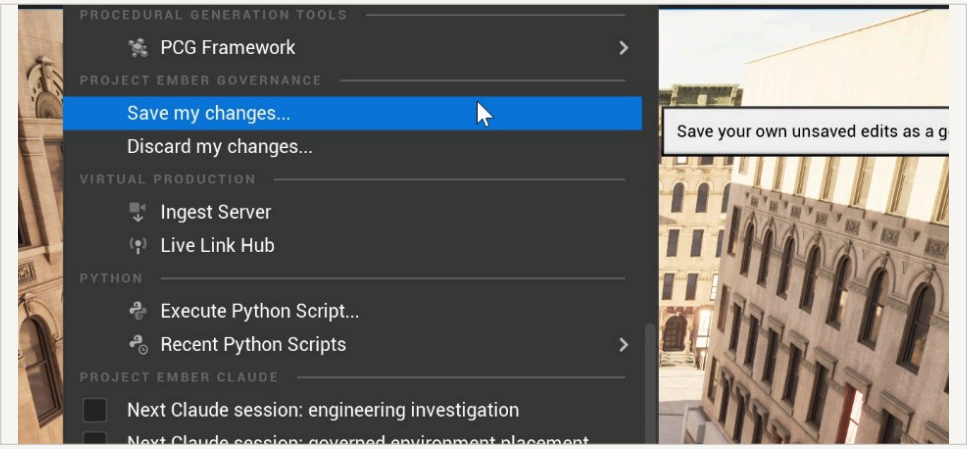
I created and initially implemented the foundational game code and definitions, then designed custom tooling and AI-assisted workflows as the project grew. Later work used a combination of direct implementation, engineering direction, assisted development, and in-editor validation.

CORE TECHNICAL AREAS

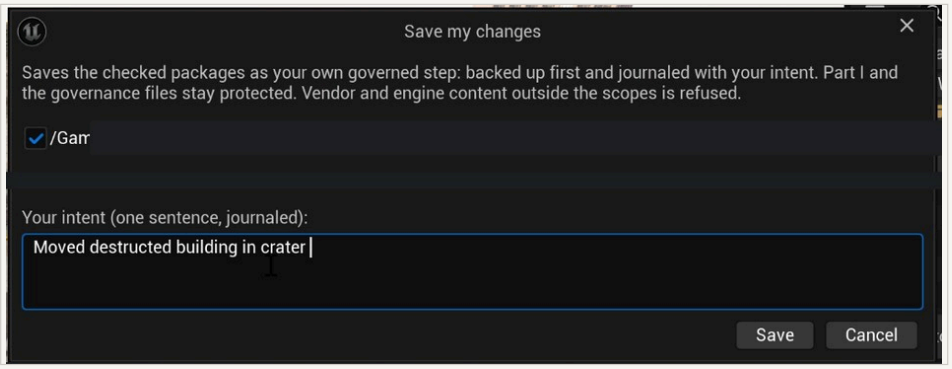
C++ gameplay components • modular framework design • editor tooling • governed development operations • world and environment studies • technical validation

From intent to a recorded change

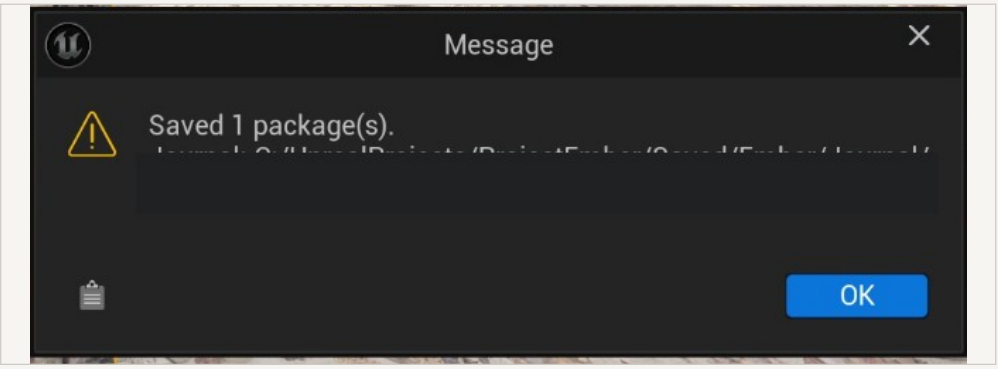
A demonstrated custom Unreal Editor workflow, not a conceptual interface.



01 EDITOR ACTION AVAILABLE IN UNREAL



02 RECORD INTENT AND PACKAGE SCOPE



03 SAVE CONFIRMED ONE PACKAGE SAVED JOURNAL RECORDED

COGNITIVE ENGINEERING

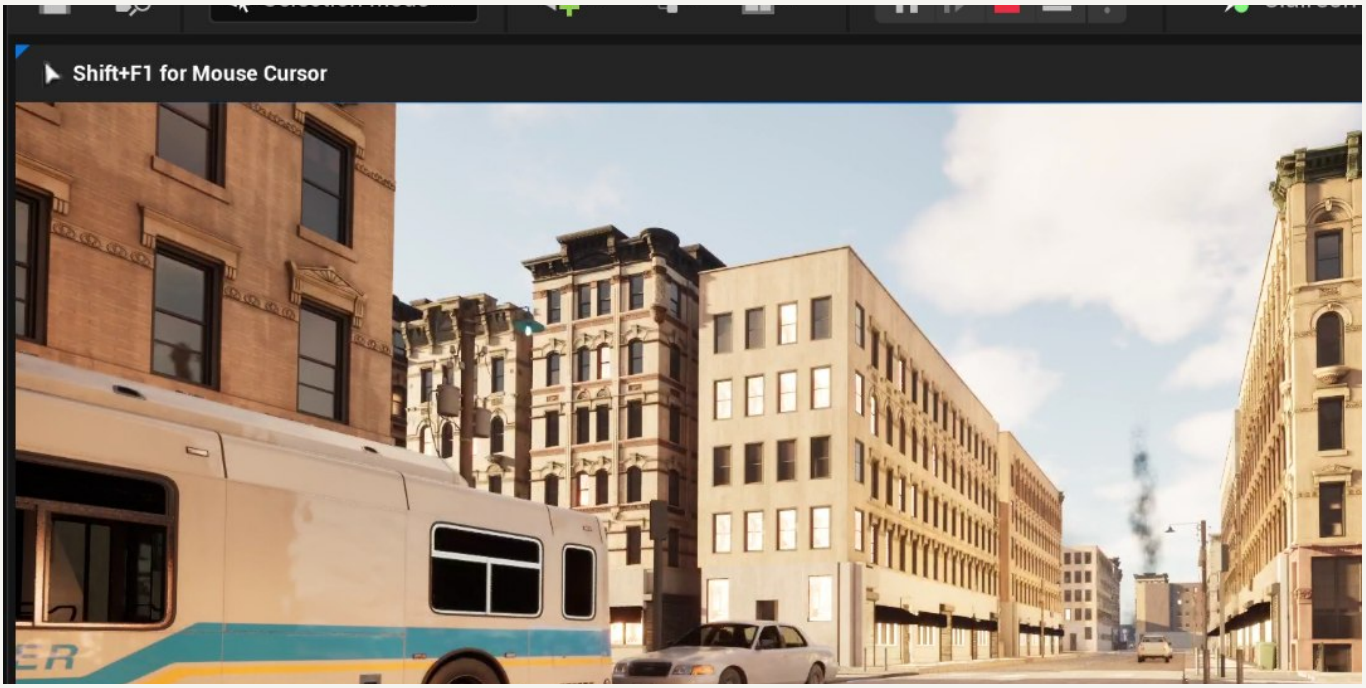
The custom plugin and bridge explore how AI-supported development can operate with explicit engineering context, scoped authority, validation, and visible outcomes. The editor-facing save workflow lets a human select a change, state its purpose, and receive a saved-package confirmation and journal reference.

EVIDENCE BOUNDARY

These captures demonstrate the interface and a completed save. They do not, by themselves, prove every authorization, backup, or recovery guarantee. Sensitive local paths have been obscured.

The systems serve the game

World construction and event studies are tested within Unreal, not only described in documents.



PLAY IN EDITOR / LOW CAMERA VIEW DURING THE STREET EVENT



OBSERVED ENVIRONMENT / BASE



OBSERVED ENVIRONMENT / POST

APPLIED DEVELOPMENT

The Part I study demonstrates a bounded playable sequence: ordinary street activity, escalating wrongness, impact, player recovery, and a rescue decision. Iterative environment work develops the urban context around the mechanics while protecting successful behavior.

The paired alley captures show scene-state differences from a matched viewpoint. They are not presented as proof that every visual change was procedurally generated or caused by runtime destruction.

From observation to decisions

Snapshot research turns complex world state into a bounded next production action.



LARGER POST-EVENT WORLD STUDY / CRATER AND CITY EDGE

SNAPSHOT TECHNOLOGY

I investigated Unreal world representations through comparative reference projects and read-only extraction of actors, components, instances, and spatial relationships. The work separates what is observed from what is intended, then identifies matches, gaps, conflicts, and unknowns.

THE PRACTICAL DECISION

For Part I, the research concluded that the mechanical sequence should be preserved while the surrounding Auburn urban environment was expanded. That focused subsequent work on city continuity, architecture, event framing, and readable aftermath rather than rebuilding already-tested mechanics.

STATUS AND ATTRIBUTION

Independent, ongoing project; development studies and selected tooling have been demonstrated. Broader world-evolution systems and general-purpose engineering automation remain research or staged capabilities. Unreal Engine and third-party assets retain their respective ownership; the portfolio represents my design, code, integration, research, and production contributions, not authorship of every asset.